frequencies of the hypothesized distribution. The test statistic is of the form $\chi^2 = \sum \left((o_i - e_i)^2 / e_i \right)$, where o_i and e_i are the observed and expected frequencies of occurrence of the measure, respectively.

For many of the tests in this test suite, the assumption has been made that the size of the sequence length, n, is large (of the order 10^3 to 10^7). For such large sample sizes of n, asymptotic reference distributions have been derived and applied to carry out the tests. Most of the tests are applicable for smaller values of n. However, if used for smaller values of n, the asymptotic reference distributions would be inappropriate and would need to be replaced by exact distributions that would commonly be difficult to compute.

Note: For many of the examples throughout Section 2, small sample sizes are used for illustrative purposes only, e.g., $n = 10$. The normal approximation is not really applicable for these examples.

2.1 Frequency (Monobit) Test

2.1.1 Test Purpose

The focus of the test is the proportion of zeroes and ones for the entire sequence. The purpose of this test is to determine whether the number of ones and zeros in a sequence are approximately the same as would be expected for a truly random sequence. The test assesses the closeness of the fraction of ones to ½, that is, the number of ones and zeroes in a sequence should be about the same. All subsequent tests depend on the passing of this test.

2.1.2 Function Call

Frequency(n), where:

n The length of the bit string.

Additional input used by the function, but supplied by the testing code:

ε The sequence of bits as generated by the RNG or PRNG being tested; this exists as a global structure at the time of the function call; $\varepsilon = \varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$.

2.1.3 Test Statistic and Reference Distribution

s_{obs}: The absolute value of the sum of the X_i (where, $X_i = 2\varepsilon - 1 = \pm 1$) in the sequence divided by the square root of the length of the sequence.

The reference distribution for the test statistic is half normal (for large n). (Note: If z (where $z = s_{obs} / \sqrt{2}$; see Section 3.1) is distributed as normal, then $|z|$ is distributed as half normal.) If the sequence is random, then the plus and minus ones will tend to cancel one another out so that the test statistic will be about 0. If there are too many ones or too many zeroes, then the test statistic will tend to be larger than zero.

2.1.4 Test Description

(1) Conversion to ±1: The zeros and ones of the input sequence (ε) are converted to values of –1 and +1 and are added together to produce $S_n = X_1 + X_2 + \cdots + X_n$, where $X_i = 2\varepsilon_i - 1$.

2.2 Frequency Test within a Block

2.2.1 Test Purpose

The focus of the test is the proportion of ones within M-bit blocks. The purpose of this test is to determine whether the frequency of ones in an M-bit block is approximately M/2, as would be expected under an assumption of randomness. For block size M=1, this test degenerates to test 1, the Frequency (Monobit) test.

2.2.2 Function Call

BlockFrequency(*M,n*), where:

M The length of each block.

n The length of the bit string.

Additional input used by the function, but supplied by the testing code:

ε The sequence of bits as generated by the RNG or PRNG being tested; this exists as a global structure at the time of the function call; $\varepsilon = \varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$.

2.2.3 Test Statistic and Reference Distribution

$\chi^2(obs)$: A measure of how well the observed proportion of ones within a given M-bit block match the expected proportion (1/2).

The reference distribution for the test statistic is a χ^2 distribution.

2.2.4 Test Description

(1) Partition the input sequence into $N = \left\lfloor \frac{n}{M} \right\rfloor$ non-overlapping blocks. Discard any unused bits.

For example, if $n = 10$, $M = 3$ and $\varepsilon = 0110011010$, 3 blocks ($N = 3$) would be created, consisting of *011, 001 and 101*. The final *0* would be discarded.

(2) Determine the proportion π_i of ones in each M-bit block using the equation $\pi_i = \frac{\sum_{j=1}^{M} \varepsilon_{(i-1)M+j}}{M}$, for $1 \le i \le N$.

For the example in this section, $\pi_1 = 2/3$, $\pi_2 = 1/3$, and $\pi_3 = 2/3$.

(3) Compute the χ^2 statistic: $\chi^2(obs) = 4M \sum_{i=1}^{N} (\pi_i - 1/2)^2$.

For the example in this section, $\chi^2(obs) = 4 \times 3 \times \left(\left(\frac{2}{3} - \frac{1}{2}\right)^2 + \left(\frac{1}{3} - \frac{1}{2}\right)^2 + \left(\frac{2}{3} - \frac{1}{2}\right)^2 \right) = 1$.

For the example, *P-value* = $\boldsymbol{erfc}\left(\frac{7-\left(2\bullet 10\bullet\frac{3}{5}\left(1-\frac{3}{5}\right)\right)}{2\bullet\sqrt{2\bullet 10}\bullet\frac{3}{5}\bullet\left(1-\frac{3}{5}\right)}\right)$ = 0.147232.

2.3.5 Decision Rule (at the 1% Level)

If the computed *P-value* is < 0.01, then conclude that the sequence is non-random. Otherwise, conclude that the sequence is random.

2.3.6 Conclusion and Interpretation of Results

Since the *P-value* obtained in step 4 of Section 2.3.4 is ≥ 0.01 (i.e., *P-value* = 0.147232), the conclusion is that the sequence is random.

Note that a large value for $V_n(obs)$ would have indicated an oscillation in the string which is too fast; a small value would have indicated that the oscillation is too slow. (An oscillation is considered to be a change from a one to a zero or vice versa.) A fast oscillation occurs when there are a lot of changes, e.g., 010101010 oscillates with every bit. A stream with a slow oscillation has fewer runs than would be expected in a random sequence, e.g., a sequence containing 100 ones, followed by 73 zeroes, followed by 127 ones (a total of 300 bits) would have only three runs, whereas 150 runs would be expected.

2.3.7 Input Size Recommendation

It is recommended that each sequence to be tested consist of a minimum of 100 bits (i.e., $n \geq 100$).

2.3.8 Example

(input) ε = 1100100100001111110110101010001000100001011010001100001000110100110001001100011001100010100001011100 0

(input) n = 100

(input) τ = 0.02

(processing) π = 0.42

(processing) $V_n(obs)$ = 52

(output) *P-value* = 0.500798

(conclusion) Since *P-value* ≥ *0.01*, accept the sequence as random.

2.4 Test for the Longest Run of Ones in a Block

2.4.1 Test Purpose

The focus of the test is the longest run of ones within M-bit blocks. The purpose of this test is to determine whether the length of the longest run of ones within the tested sequence is consistent with the length of the longest run of ones that would be expected in a random sequence. Note that an irregularity in the expected length of the longest run of ones implies that there is also an irregularity in the expected length of the longest run of zeroes. Therefore, only a test for ones is necessary. See Section 4.4.

01001101	(2)	01010001	(1)
00010011	(2)	11010110	(2)
10000000	(1)	11010111	(3)
11001100	(2)	11100110	(3)
11011000	(2)	10110010	(2)

(processing) $\nu_0 = 4;\ \nu_1 = 9;\ \nu_2 = 3;\ \nu_4 = 0;\ \chi^2 = 4.882457$

(output) *P-value = 0.180609*

(conclusion) Since the *P-value* is ≥ 0.01, accept the sequence as random.

2.5 Binary Matrix Rank Test

2.5.1 Test Purpose

The focus of the test is the rank of disjoint sub-matrices of the entire sequence. The purpose of this test is to check for linear dependence among fixed length substrings of the original sequence. Note that this test also appears in the DIEHARD battery of tests [7].

2.5.2 Function Call

Rank(*n*), where:

n The length of the bit string.

Additional input used by the function supplied by the testing code:

The sequence of bits as generated by the RNG or PRNG being tested; this exists as a global structure at the time of the function call; $\varepsilon = \varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$.

M The number of rows in each matrix. For the test suite, *M* has been set to 32. If other values of *M* are used, new approximations need to be computed.

Q The number of columns in each matrix. For the test suite, *Q* has been set to 32. If other values of *Q* are used, new approximations need to be computed.

2.5.3 Test Statistic and Reference Distribution

$\chi^2(obs)$: A measure of how well the observed number of ranks of various orders match the expected number of ranks under an assumption of randomness.

The reference distribution for the test statistic is a χ^2 distribution.

2.5.4 Test Description

(1) Sequentially divide the sequence into $M \bullet Q$-bit disjoint blocks; there will exist $N = \left\lfloor \frac{n}{MQ} \right\rfloor$ such blocks. Discarded bits will be reported as not being used in the computation within each block. Collect the $M \bullet Q$ bit segments into *M* by *Q* matrices. Each row of the matrix is filled with successive *Q*-bit blocks of the original sequence ε.

2.9 Maurer's "Universal Statistical" Test

2.9.1 Test Purpose

The focus of this test is the number of bits between matching patterns (a measure that is related to the length of a compressed sequence). The purpose of the test is to detect whether or not the sequence can be significantly compressed without loss of information. A significantly compressible sequence is considered to be non-random.

2.9.2 Function Call

Universal(*L, Q, n*), where

L The length of each block. Note: the use of L as the block size is not consistent with the block size notation (*M*) used for the other tests. However, the use of L as the block size was specified in the original source of Maurer's test.

Q The number of blocks in the initialization sequence.

n The length of the bit string.

Additional input used by the function, but supplied by the testing code:

ε The sequence of bits as generated by the RNG or PRNG being tested; this exists as a global structure at the time of the function call; $\varepsilon = \varepsilon_1, \varepsilon_2, \ldots, \varepsilon_n$.

2.9.3 Test Statistic and Reference Distribution

f_n : The sum of the log_2 distances between matching *L*-bit templates, i.e., the sum of the number of digits in the distance between *L*-bit templates.

The reference distribution for the test statistic is the half-normal distribution (a one-sided variant of the normal distribution) as is also the case for the Frequency test in Section 2.1.

2.9.4 Test Description

(1) The *n*-bit sequence (ε) is partitioned into two segments: an initialization segment consisting of *Q* *L*-bit non-overlapping blocks, and a test segment consisting of *K* *L*-bit non-overlapping blocks. Bits remaining at the end of the sequence that do not form a complete *L*-bit block are discarded. The first *Q* blocks are used to initialize the test. The remaining *K* blocks are the test blocks ($K = \lfloor n/L \rfloor - Q$).

